

YUKAI SONG

Pittsburgh, PA • yus190@pitt.edu • +1 917-251-8302 • [Yukai's Website](#)

Full-time internship availability: May–September 2027

SUMMARY

Ph.D. candidate in Electrical & Computer Engineering focused on reliable and efficient machine learning for content risk detection and LLM/agent applications. Experience spans agentic RAG evaluation, uncertainty-aware NLP, computer vision, and model compression, with research contributions in selective escalation, retrieval failure analysis, and safety-aware model selection.

EDUCATION

University of Pittsburgh — Ph.D. in Electrical & Computer Engineering | GPA: 4.0

Pittsburgh, PA | Aug 2023 – Present | Expected graduation: 2027

Advisors: Prof. Jingtong Hu and Prof. Zhi-Hong Mao | Proposal defended: Jan 2026

Columbia University — M.S. in Electrical Engineering | GPA: 3.96

New York, NY | Sep 2021 – Feb 2023

University of Electronic Science and Technology of China — B.S. in Electronic Information Engineering

Glasgow College, Chengdu, China | Sep 2017 – Jun 2021 | GPA: 3.92; Rank: 1/257

National Scholarship (Top 2%) | First Class Honours, joint program with University of Glasgow

RESEARCH & PROJECT EXPERIENCE

Selective Escalation for Social-Media Suicide Risk Detection

Apr 2025 – Present

First-author research; manuscript under review at JMIR AI | [Preprint](#)

- Designed a two-stage NLP framework that uses confidence- and length-based routing to handle explicit cases with fine-tuned BERT and escalate ambiguous posts to multi-perspective LLM voting or a psychological-feature ensemble.
- Kept 67.6% of 23,200 held-out Reddit posts in the BERT stage, limiting second-stage processing to 32.4%; the feature-ensemble variant achieved 97.99% F1 versus 97.41% for BERT.
- Evaluated cross-domain behavior on synthetic implicit-risk posts, analyzing recall, F1, and routing coverage to characterize difficult content and the limits of confidence-based escalation.

DSH DocsQA: Agentic RAG Evaluation & Failure Analysis

Aug 2026 – Present

Research contributor — evaluation design and results analysis | [GitHub](#)

- Authored a merged research roadmap for a documentation-QA benchmark now spanning 467 questions and 4,860 pages across four products; identified sparse citation-label limitations and defined separate evaluation tracks for retrieval, evidence grounding, and agent efficiency.
- Diagnosed candidate-generation and ranking failures across five retrieval baselines and 2,335 evaluations: native-link expansion reduced candidate misses by 26.5% without increasing total top-10 citation recovery; the tested reranker lowered Recall@10 by 9.06 percentage points.
- Analyzed a reported 361-question agent comparison: hybrid retrieval used 69.0% fewer tokens per question and improved weighted answer-aspect coverage by 7.58 percentage points versus filesystem search, while exposing a higher unsupported-claim rate.

Intelligent Ping Pong Coach

Dec 2022 – Jun 2025

Independent project — vision/sensor coaching and conversational retrieval | [Vision demo](#) | [Custom GPT](#)

- Built real-time posture and stroke analysis with MediaPipe, OpenCV, and IMU sensor fusion to provide personalized coaching feedback; the public demo received 4,100+ views.
- Developed a RAG-powered Custom GPT that retrieves professional training knowledge for personalized, multi-turn guidance; the conversational coach recorded 100+ chats.

RESEARCH & PROJECT EXPERIENCE (CONTINUED)

Reliable ML for EEG-Based Sleep Stage Classification

Jan 2026 – Present

Dissertation research

- Designed and implemented a two-stage reliability framework using cross-model label disagreement and distributional divergence to identify high-confidence errors and route high-risk predictions for further processing.
- Completed an initial 20-fold leave-one-subject-out evaluation on Sleep-EDF-20: identified 1,923 high-confidence errors and achieved an accepted-sample high-confidence-error rate of 1.62%, with 18.8% of samples escalated.
- Extending the initial framework toward probability-calibration audits, subject-level risk estimation, and risk-controlled selective prediction under held-out-subject evaluation.

Controller-Aware Perception Network Pruning for CPS Safety

Sep 2023 – Jul 2025

- Co-developed a YOLOv8 pruning framework that incorporates downstream control requirements; used reachability analysis, maximum reachable-set diameter, and regression-based scaling laws for safety-aware model selection.
- Enabled selection of the smallest pruned model satisfying evaluated safety constraints; achieved up to 148.33× speedup in pruning search on distance-estimation tasks while meeting hardware and safety objectives.

Deep Learning for Contactless Human Motion Detection

Sep 2021 – Dec 2022

- Built signal-processing and deep-learning pipelines for RF-based human motion classification, using Butterworth filtering and artifact removal before neural-network inference.
- Compared neural-network and PCA configurations on 700 motion samples: RNN with PCA achieved 99.1% accuracy versus 93.8% for the basic neural-network baseline; published in Scientific Reports.

SELECTED PEER-REVIEWED PUBLICATIONS

- Chen, J., **Song, Y.**, Xu, S., *et al.* (2026). Controller-Aware Perception Neural Network Pruning for CPS Safety Optimization. *ACM Transactions on Cyber-Physical Systems*.
- Tang, Y., **Song, Y.**, Elango, N., *et al.* (2024). CHEF: A Framework for Deploying Heterogeneous Models on Clusters With Heterogeneous FPGAs. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 43(11), 3937–3948.
- **Song, Y.**, Taylor, W., Ge, Y., *et al.* (2022). Evaluation of deep learning models in contactless human motion detection system for next generation healthcare. *Scientific Reports*, 12, 21592.
- **Song, Y.**, Taylor, W., Ge, Y., *et al.* (2021). Design and Implementation of a Contactless AI-enabled Human Motion Detection System for Next-Generation Healthcare. In *IEEE International Conference on Smart Internet of Things (SmartIoT)*, pp. 112–119. **(Best Paper Award)**

MANUSCRIPT UNDER REVIEW

- **Song, Y.**, Zhou, P., Escobar-Viera, C. G., *et al.* A Two-Stage Selective Escalation Framework for Robust Suicide Risk Detection on Social Media: Machine Learning Study. Under review at *JMIR AI*.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, CUDA

ML & Vision: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, MediaPipe

Methods: RAG/LLM evaluation, retrieval analysis, uncertainty quantification, selective prediction, model pruning

Tools: Git, Docker, Linux, Jupyter, AWS